



24th Linux SIG Workshop

Disaster Recovery: Be Prepared

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Agenda

- ◆ Disaster Recovery Definition
- ◆ High Availability vs. Disaster Recovery
- ◆ Some more definitions
- ◆ Disaster Recovery Process/Planning
- ◆ Make CD-ROM Recovery (mkCDrec)
- ◆ The mkCDrec tool explained
See: <http://mkcdrec.ota.be/>

Disaster Recovery Definition

- ◆ Disaster recovery is concentrated on getting the equipment up and running after a catastrophe
- ◆ Disaster recovery also means an interruption of operations
 - ◆ IT infrastructure may be damaged
 - ◆ IT infrastructure may be not accessible
 - ◆ IT personnel may be out of order
- ◆ Disaster recovery is a low end solution

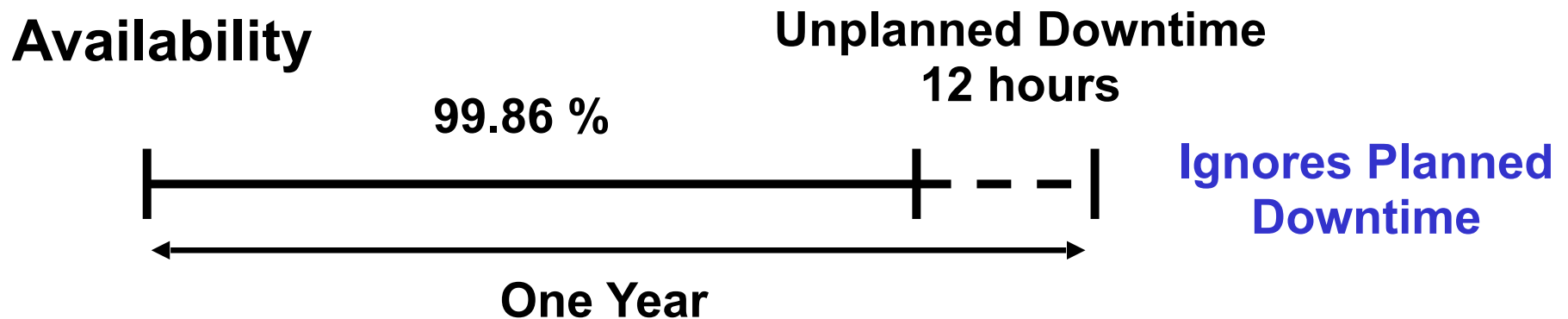
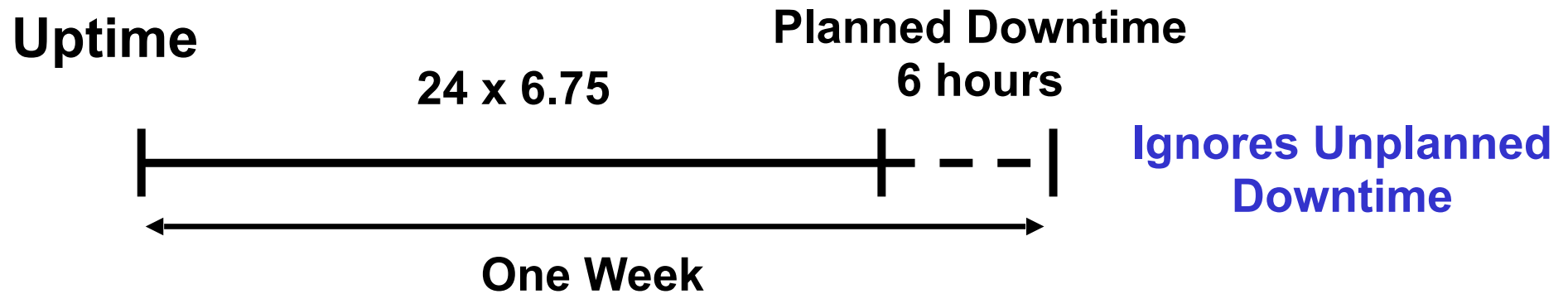
Is Disaster Recovery Planning needed?

- ◆ A study by the University of Texas found that companies after a catastrophic data loss
 - ◆ 43% never reopened
 - ◆ 51% closed after 2 years
 - ◆ **6% survived**
- ◆ World Trade Bombing as an example:
 - ◆ 50% of companies without Disaster Recovery Plan were out of business within 2 years

High Availability vs. Disaster Recovery

- ◆ **High Availability** is designing the infrastructure to prevent loss of IT service to users
- ◆ **Highly available computing** uses computer systems which are *designed and managed to operate with only a small amount of planned and unplanned downtime.*
- ◆ **Disaster Recovery Plan** has to be made and practiced to rebuild the damaged IT infrastructure
 - ◆ IT people are not that easy to replace!

Terminology



What is Availability?

- ◆ Formula:

$$\frac{(\text{Total elapsed time} - \sum \text{ of Inoperative Time})}{\text{Total Elapsed Time}}$$

- ◆ % Availability =

- ◆ 12 * 5 * 52 with 99 % Availability means

- ◆ 3088 hours available with max. 32 hours allowed downtime

- ◆ 5640 hours free for maintenance purposes



Being Available

? Uptime

24 x 7
24 x 6.75
16 x 5

? Downtime

planned
unplanned

? Outage

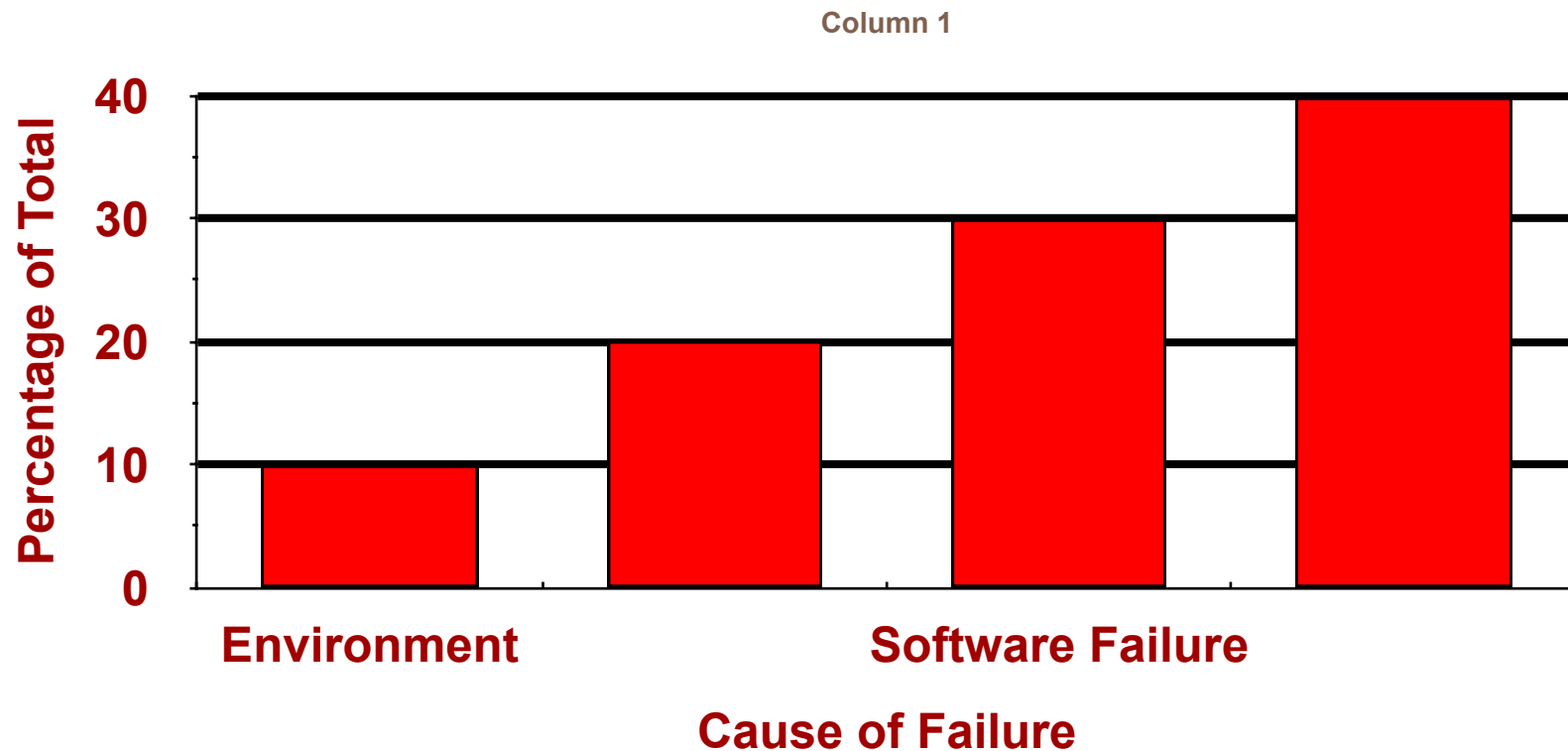
? Single Points of Failure (SPOFs)

hardware
environment

? Availability (annual)

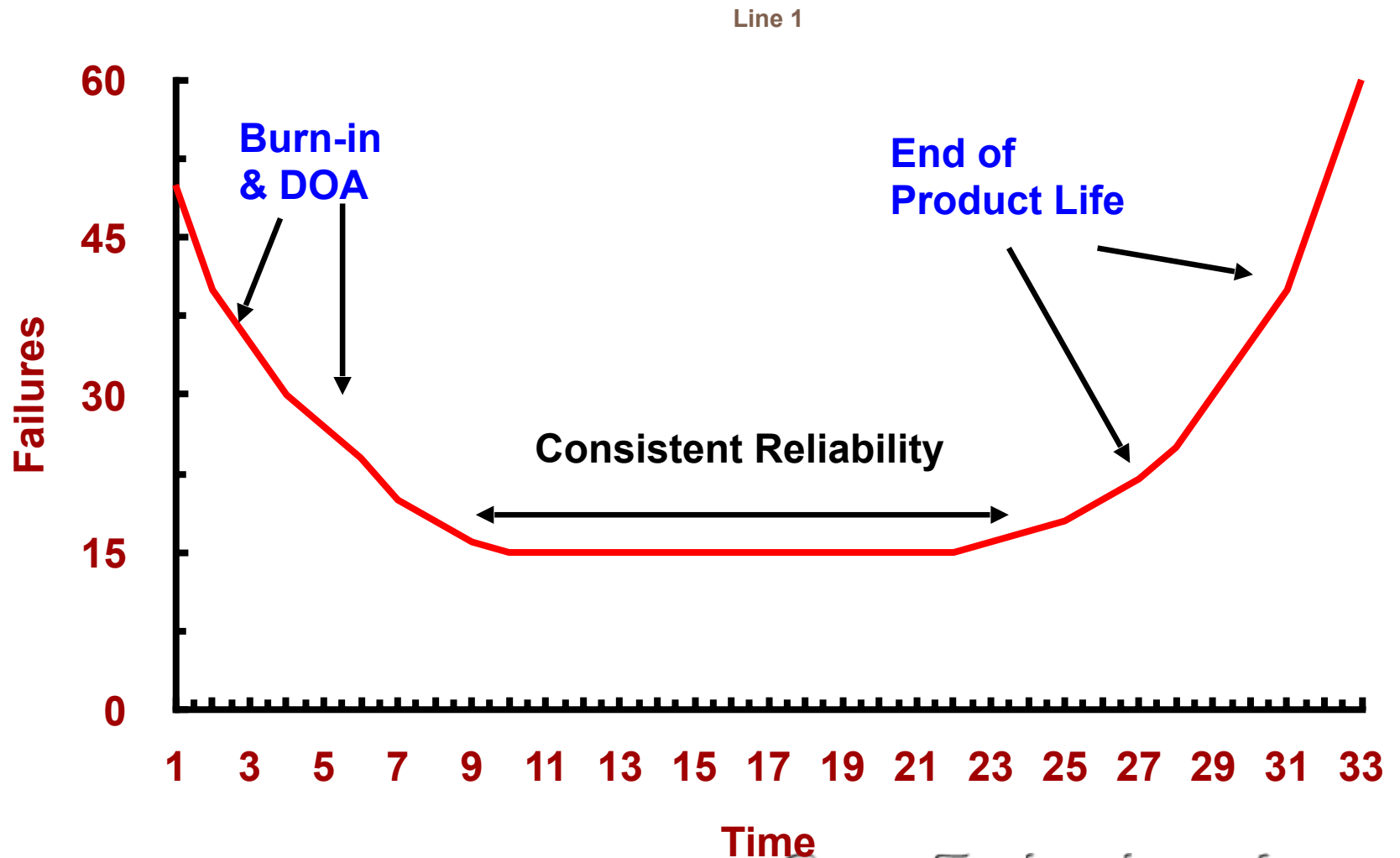
99.999	5.4	minutes
99.99	52.8	minutes
99.95	4.4	hours
99.90	8.8	hours
99.86	12.3	hours
99.82	15.8	hours
99.73	24	hours
99	3.6	days
98	7.2	days
97	10.8	days
96	14.4	days
95	18	days

Causes of System Failure



Source : Gartner Group 1993

Failure Rate Curve (Bathtub Curve)



Working towards High Availability

- ◆ Routine backups
- ◆ Routine maintenance
- ◆ Software upgrades
- ◆ Recovery following failure
 - ◆ Disaster Recovery Plan (DRP)
 - ◆ Planning for disaster
 - ◆ Preparedness

7 Steps away from DRP

- ◆ Define goals, objectives and assumptions
- ◆ Conduct a Business Impact Analysis
- ◆ Determine business recovery strategies
- ◆ Prepare and implement the plan
- ◆ Perform realistic testing
- ◆ Update and refine the plan
- ◆ Be prepared

<http://www1.dlittape.com/proveit/steps/plan/drp/toolkit.asp>

Tools on the Market

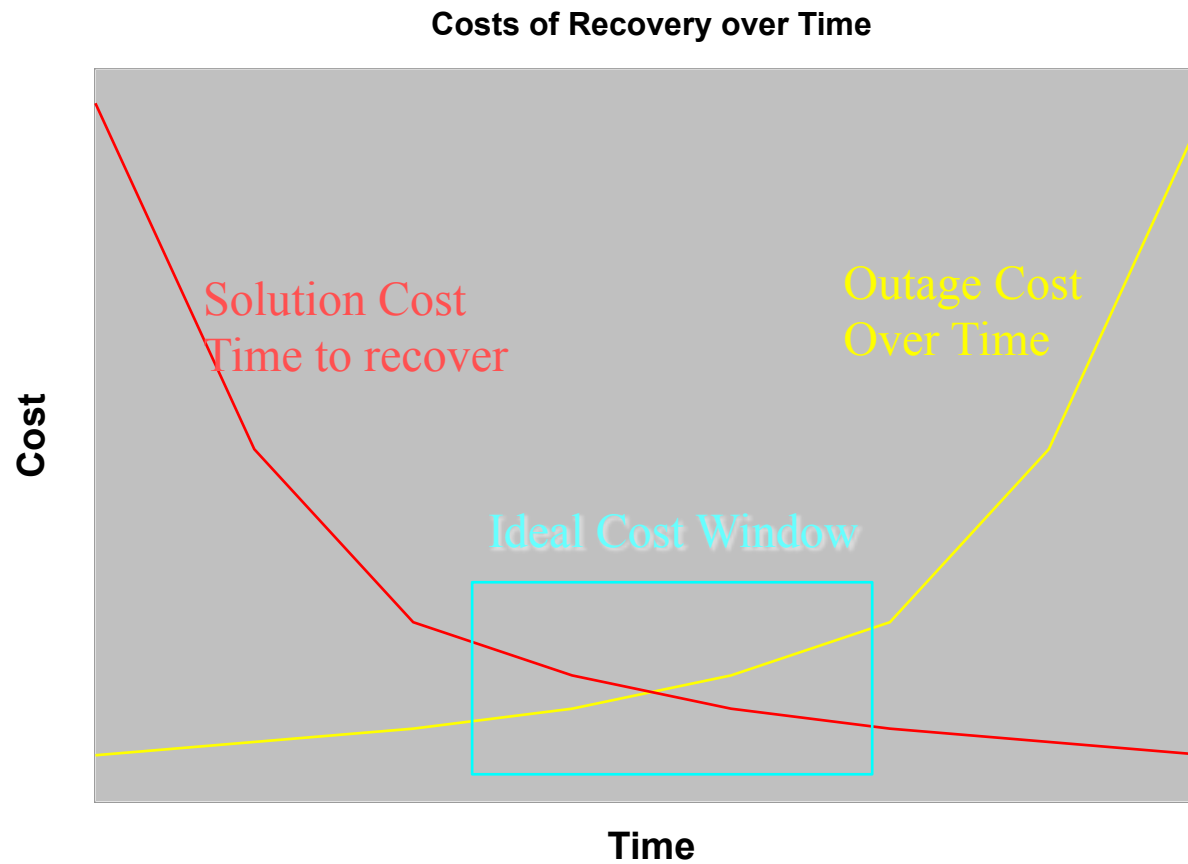
◆ Windows:

- ◆ Ghost: make an image of your system to disk
 - ◆ Needs plenty of free disk space
- ◆ HP One-Button Disaster Recovery: bootable tape
 - ◆ Needs a OBDR aware tape drive (of HP)

◆ Linux:

- ◆ Crash Recovery Utility (from EST Inc)
 - ◆ Needs BRU from EST Inc (commercial)
 - ◆ Needs OBDR aware tape drive

Cost of Downtime



Back to Reality

- ◆ How long can you be out of the air?
- ◆ What is the business impact?
- ◆ Can you loose customers?
- ◆ Do customers run to the competition?
 - ◆ Will customers come back?
 - ◆ What do you have to do to get them back?
- ◆ What are the hidden costs?
- ◆ Do I have a support contract?

The lesson: Be Prepared

- ◆ Make back-ups!
- ◆ Make a Disaster Recovery Plan/Procedure
 - ◆ Exercise it
 - ◆ Improve it
- ◆ Look for tools and test them on usefulness
- ◆ Keep a documentation map of your systems

Rescue floppies / CD-roms

- ◆ <http://freshmeat.net/search/?q=rescue>

- ◆ 23 hits

- ◆ <http://freshmeat.net/search/?q=boot>

- ◆ 25 hits

- ◆ Rescue floppies:

- ◆ RamFloppy, Yard, Resque_disk, tomsrtbt

- ◆ Rescue CD-roms:

- ◆ Repairlix, Superrescue, Timos_rescue_CD_set

Make CD-ROM Recovery

Mkcdrec

- ◆ Did not found where I was looking for...
- ◆ Make CD-ROM recovery (mkCDrec)
 - ◆ Is a rescue CD-rom
 - ◆ CD-rom is bootable (a boot-floppy is optional)
 - ◆ May contain the backups of the original system on
 - ◆ CD-rom, local spare disk, across NFS, tape
 - ◆ Start_recover.sh will (semi-)automatically re-install the system

mkCDrec Features

Mkcdrec

- ◆ El Torito bootable Rescue CD-ROM
 - ◆ Contains the current kernel + loadable modules
 - ◆ Basic tools to recover from failures
- ◆ Disaster recovery CD-ROM
 - ◆ Backup of hard disk partitions on CD-ROM
 - ◆ Backup to other (local disk) or NFS disk
 - ◆ Backup to tape
- ◆ Keep it simple

mkCDrec prerequisites

Mkcdrec

- ◆ A recent Linux kernel 2.2.* and higher
- ◆ A Linux source tree to find out:
 - ◆ RAM disk support: **Y**
 - Default RAM disk size **[4096]**
 - Initial RAM disk (initrd) support **Y**
 - Loopback device support **Y (or m)**
 - ISO9660 CDRom filesystem **Y**
 - Microsoft Joliet CDRom ext. **Y**
- ◆ Mkisofs-1.13 (can be downloaded from OTA)

mkCDrec includes

Mkcdrec

- ◆ Busybox-0.47 combines lots of utilities in a single binary
 - ◆ 70 utilities (excluded tar, prefer GNU tar)
 - ◆ tty[1-4] use /bin/sh (after askfirst = keypress)
- ◆ Tinylogin-0.78 is included (tty5)
- ◆ Tty6 contains the output of syslogd
- ◆ Sfdisk-3.07 (simple fdisk non-interactive mode)
- ◆ Syslinux-1.49 to make CD bootable (lilo alike)

mkCDrec Shortcomings

Mkcdrec

- ◆ SCSI support has to be in the kernel to mount the root image from a **SCSI CDROM** automatically
 - ◆ SCSI modules are supported afterwards when the root image has become active
- ◆ Must have the linux source tree for some initial checks
 - ◆ RAM disk, RAM initrd and loopback device support
- ◆ A linux kernel rebuild is probably necessary

Configuration File

Mkcdrec

◆ Config.sh

- ◆ `PATH=${PATH}:${MKCDREC_DIR}/bin`
- ◆ `MKCDREC_DIR=`pwd`` # where sources are
- ◆ `ISOFS_DIR=/some/path` # where temp. files go for CD
- ◆ `TAPE_DEV=/dev/nst0` # no-rewinding (if available)
- ◆ `BUSYBOX_DIR=${MKCDREC_DIR}/busybox-0.47`
- ◆ `TINYLOGIN_DIR=${MKCDREC_DIR}/tinylogin-0.78`
- ◆ `SFDISK_DIR=${MKCDREC_DIR}/sfdisk-3.07`
- ◆ `SCSI_MODULES="aha152x_cs"`

Configuration File (cont'd)

Mkcdrec

- ◆ OTHER_MODULES="lockd"
- ◆ MKDIR_LIST="....."
- ◆ ETC_FILES="..."
- ◆ SBIN_BINARIES="..."
- ◆ BIN_BINARIES="..."
- ◆ USRBIN_BINARIES="..."
- ◆ LINKS="/bin=sh=bash"

Makefile

Mkcdrec

- ◆ VPATH : do not remove ./bin as it contains
 - ◆ mformat, mkisofs, sfdisk, syslinux
- ◆ SYSLINUX_DIR, BUSYBOX_DIR, TINYLOGIN_DIR & SFDISK_DIR
 - ◆ Should point to the same paths as in “Config.sh”
 - ◆ These 4 applications can be upgraded...
- ◆ CDREC_ISO_DIR=. (location of CDrec.iso)
- ◆ The rest of the “Makefile” should not be changed

The “make” process

Mkcdrec

- ◆ The Makefile does lots of things:
 - ◆ Compile (if necessary) BusyBox, TinyLogin, sfdisk, syslinux
 - ◆ Runs rd-base.sh script to make big / RAM file system
 - ◆ Runs initrd.sh script to build initial RAM file system
 - ◆ Runs bootfloppy.sh script which combines the initrd and vmlinuz kernel on a 1.44 Mb floppy image (bootfloppy.img)
 - ◆ Make all the necessary backups (tar-it.sh)
 - ◆ Make the ISO9660 file system (CDrec.iso) to boot from
- ◆ Cdrecord -v -speed=2 dev=1,0,0 CDrec.iso (manual step)

Make the backups

Mkcdrec

- ◆ Tar-it.sh checks the available space in the destination path (depends which item was selected)
 - ◆ ISOFS_DIR
 - ◆ Alternate path (another local disk or NFS disk)
 - ◆ Tape
- ◆ Script warns us if the calculated size (after gzipping) exceeds available space (compress factor is 40%).
 - ◆ 50 to 60 % compression is sometimes possible!

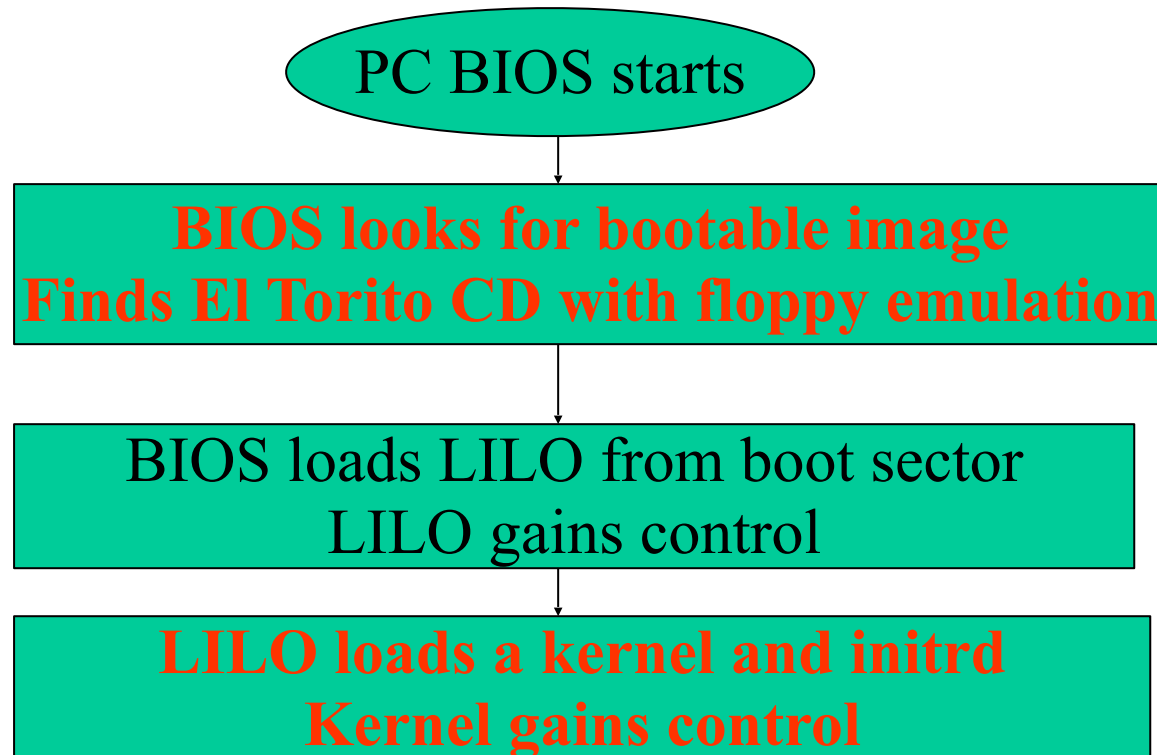
Backups Naming

Mkcdrec

- ◆ Naming convention of a backup is:
disk`partitionnr._mountpoint.tgz`
e.g. hda1._boot.tgz (_ replaces /)
- ◆ Advantage: the backups can always be restored at any given time (even without a disaster ;-)
- ◆ The log file of the backup is always kept on the CD-ROM.
 - ◆ Same naming convention (hda1._boot.log)
 - ◆ Not compressed (could easily be done...)

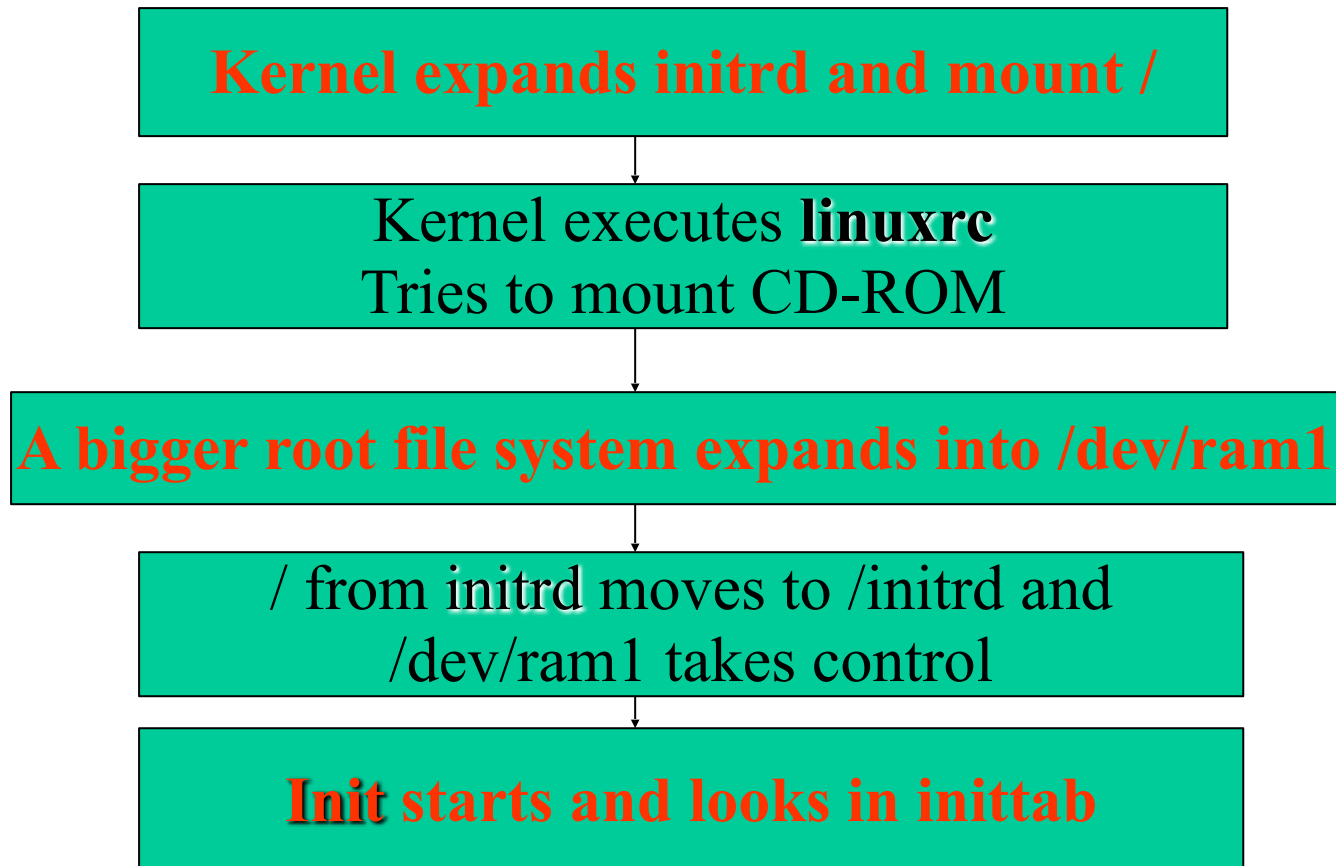
The boot process

Mkcdrec



The boot process (2)

Mkcdrec



Start-restore.sh

Mkcdrec

- ◆ Step 1: Backup or Rescue CD-ROM?
- ◆ Step 2: Location of Backups
- ◆ Step 3: Check the physical disks
- ◆ Step 4: Reformat the disks
- ◆ Step 5: Make the file systems
- ◆ Step 6: Make swap partition (if needed)
- ◆ Step 7: Restore file systems
- ◆ Step 8: LILO

http://mkCDrec.ota.be

Mkcdrec



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Make CD-ROM Recovery (mkCDrec) v0.3

by Gratien D'haese on November 22nd 2000, 18:14 EST

mkCDrec makes a bootable (El Torito) disaster recovery image (CDrec.iso), including backups of the linux system to the same CD-ROM if space permits. Otherwise, the backups can be stored on another disk, NFS disk or tape. After a disaster (disk crash or system intrusion) the system can be booted from the CD-ROM and one can restore the complete system as it was (at the time mkCDrec was run).

License: GPL

Category: Console/Administration

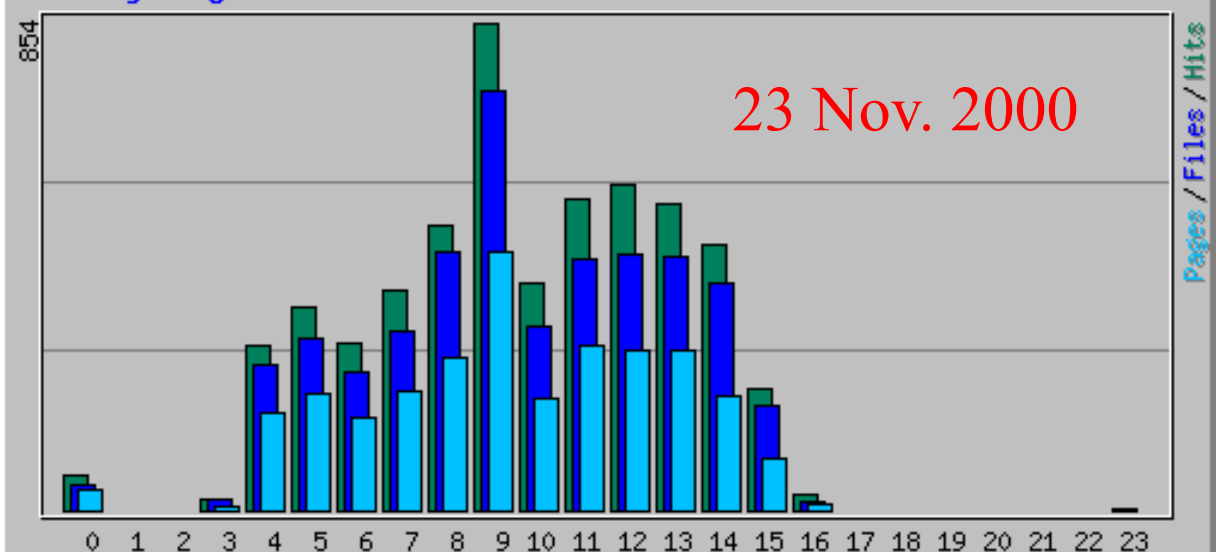
urls

Homepage: <http://>

Download: <http://>

Changelog: <http://>

Hourly usage for November 2000



References

Mkcdrec

- ◆ mkCDrec: <http://mkcdrec.ota.be/>
- ◆ OBDR: http://www.hp.com/tape/papers/obdr_wp.html
- ◆ CRU: <http://www.estinc.com/>
- ◆ Busybox: <http://www.opensource.lineo.com/>